Speedy-Splat: Fast 3D Gaussian Splatting with Sparse Pixels and Sparse Primitives

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speedysplat.github.io

Background and Motivation

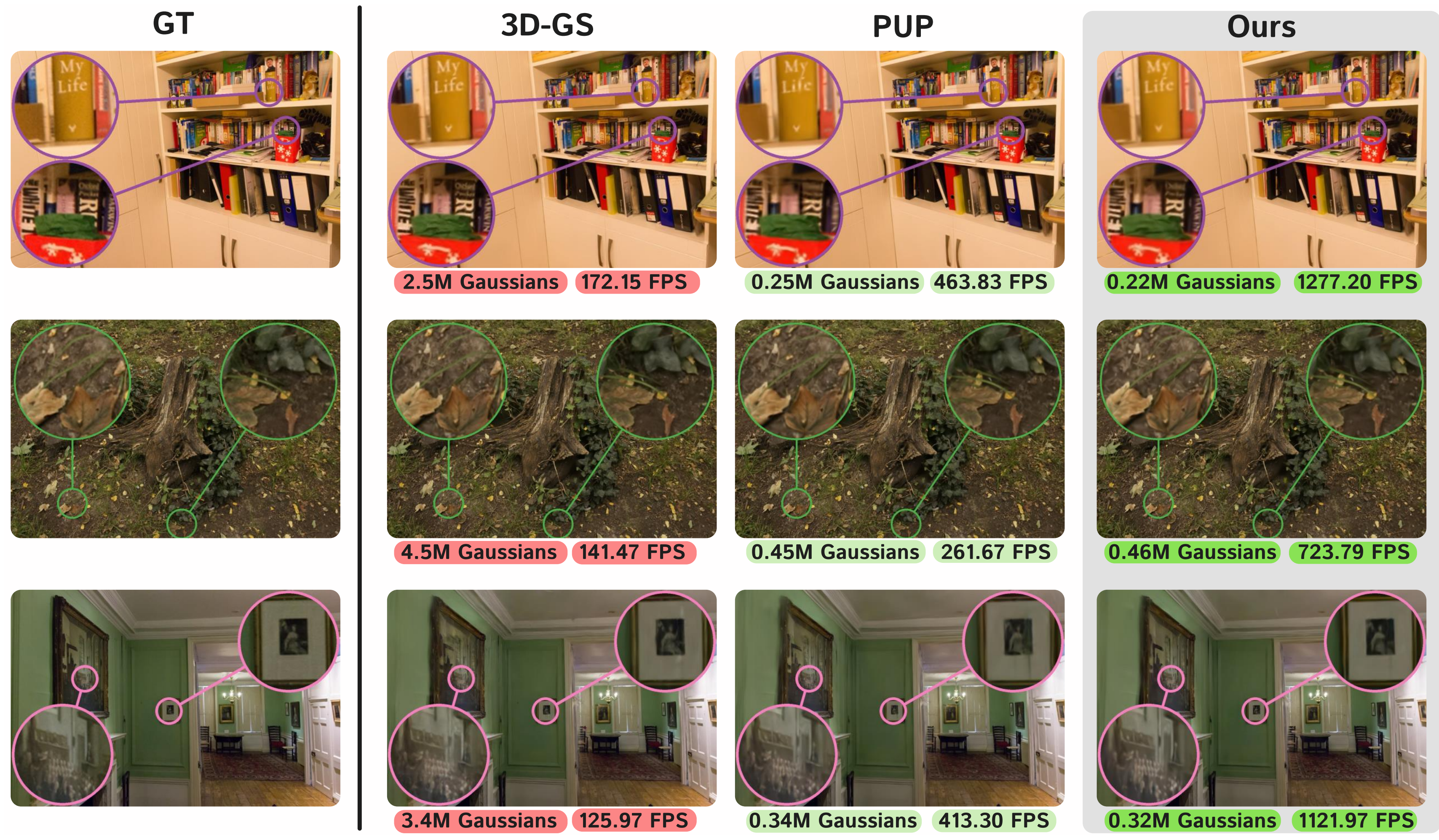
Speedy–Splat (Ours)
3D Gaussian Splatting

Speedy–Splat renders over 6 views in the time it takes 3D Gaussian Splatting to render 1

How can we accelerate the rendering speed of 3D Gaussian Splatting (3D-GS) by over 6 ×?

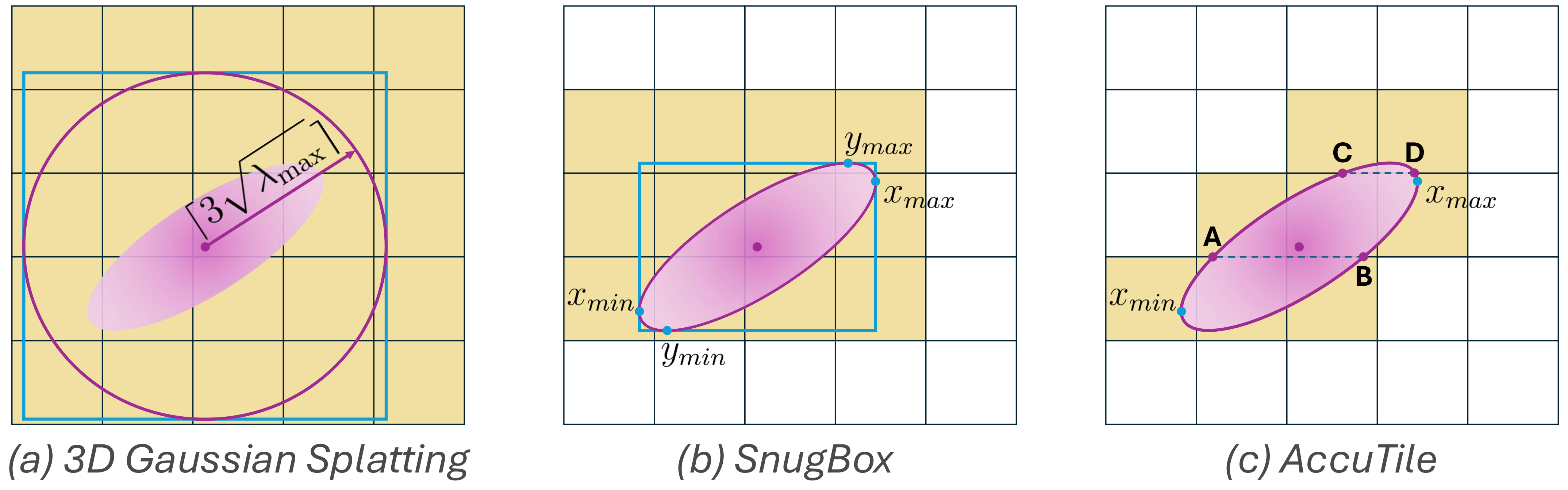
Rendering speed is primarily determined by two factors:

1. The **number of Gaussians allocated to each pixel**, and
2. The **total number of Gaussians** in the scene.


GT
3D-GS
PUP
Ours
2.5M Gaussians
172.15 FPS
0.25M Gaussians
463.83 FPS
0.22M Gaussians
1277.20 FPS
4.5M Gaussians
141.47 FPS
0.45M Gaussians
261.67 FPS
0.46M Gaussians
723.79 FPS
3.4M Gaussians
125.97 FPS
0.34M Gaussians
413.30 FPS
0.32M Gaussians
1121.97 FPS

Method

Localization


$3\sqrt{\lambda_{max}}$
y_{max}
x_{max}
x_{min}
y_{min}
C
D
A
B
x_{max}
x_{min}

(a) 3D Gaussian Splatting (b) SnugBox (c) AccuTile

Our localization algorithms reduce the number of Gaussians per pixel.

(a) 3D Gaussian Splatting overestimates Gaussian-to-tile intersections.
(b) Our **SnugBox** method finds the axis-aligned tight bounding box of the Gaussian and corresponding rectangular tile extent in constant time.
(c) Our **AccuTile** method extends SnugBox to quickly compute exact Gaussian-to-tile intersections.

SnugBox and AccuTile are **lossless** – they do not change the rendered image.

Pruning

Our pruning method reduces the total number of Gaussians by ~90%, increasing rendering and training speed while maintaining image quality. We compute a pruning score $\tilde{U}_i$ for each Gaussian $\mathcal{G}_i$ as a second order approximation of the L_2 reconstruction error:

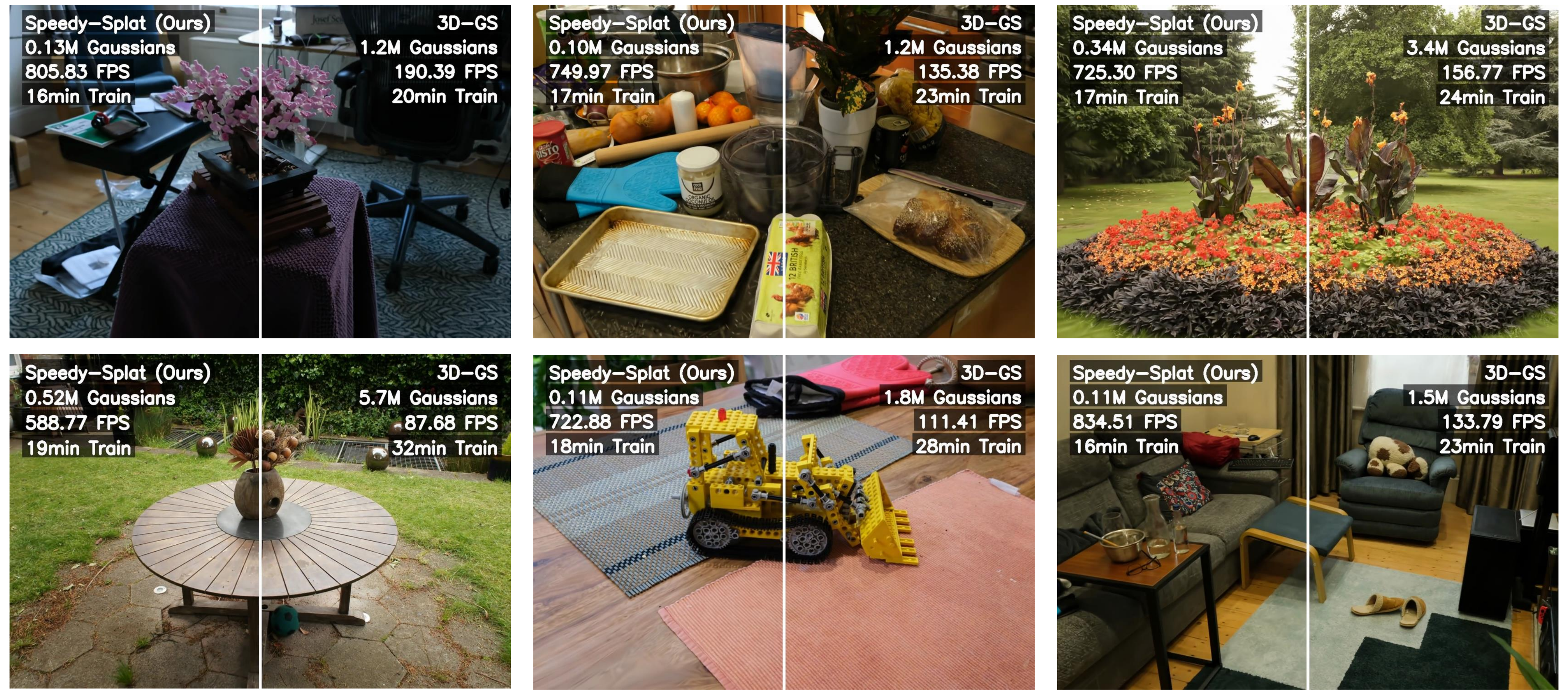
$$\nabla^2_{\mathcal{G}_i} L_2 \approx \sum_{\phi \in \mathcal{P}_{gt}} \nabla_{\mathcal{G}_i} I_{\mathcal{G}}(\phi) \nabla_{\mathcal{G}_i} I_{\mathcal{G}}(\phi)^T \approx \sum_{\phi \in \mathcal{P}_{gt}} \left(\nabla_{g_i} I_{\mathcal{G}}(\phi)\right)^2 = \tilde{U}_i,$$

where $\mathcal{P}_{gt}$ is the set of all training poses, $I_{\mathcal{G}}(\phi)$ is the rendered view for pose ϕ, and g_i is the value of the projected Gaussian in $I_{\mathcal{G}}(\phi)$.

We use this score to prune the scene during training via two modalities:

1. **Soft Pruning,** performed during the densification stage, and
2. **Hard Pruning,** performed after the densification stage.

Results


Speedy–Splat (Ours)
0.13M Gaussians
805.83 FPS
16min Train
3D–GS
1.2M Gaussians
190.39 FPS
20min Train
Speedy–Splat (Ours)
0.10M Gaussians
749.97 FPS
17min Train
3D–GS
1.2M Gaussians
135.38 FPS
23min Train
Speedy–Splat (Ours)
0.34M Gaussians
725.30 FPS
17min Train
3D–GS
3.4M Gaussians
156.77 FPS
24min Train
Speedy–Splat (Ours)
0.52M Gaussians
588.77 FPS
19min Train
3D–GS
5.7M Gaussians
87.68 FPS
32min Train
Speedy–Splat (Ours)
0.11M Gaussians
722.88 FPS
18min Train
3D–GS
1.8M Gaussians
111.41 FPS
28min Train
Speedy–Splat (Ours)
0.11M Gaussians
834.51 FPS
16min Train
3D–GS
1.5M Gaussians
133.79 FPS
23min Train

When compared to original 3D-GS, **Speedy-Splat** achieves
6.5× FPS, 10× compression, and 45% faster training.

Our **lossless** methods **boost FPS by 2×** for free.

Method	FPS↑	Comp.↑	Train↑	PSNR↑	SSIM↑	LPIPS↓
3D-GS	*1.00×*	*1.00×*	*1.00×*	27.55	0.814	0.222
EAGLES	1.51×	3.68×	1.37×	26.94	0.800	0.250
ELMGS	2.69×	5.00×	-	27.00	0.779	0.286
PUP	2.55×	8.65×	-	26.83	0.792	0.268
Mini-Splat	3.20×	6.84×	1.26×	27.34	0.822	0.217
Ours (Lossless)	1.99×	0.99×	1.10×	27.57	0.814	0.221
Ours (Full)	6.51×	10.6×	1.45×	26.94	0.782	0.296

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